Introduction To Industry 4.0 And Industrial Internet Of Things

Assignment-4

Number of questions: 15 **Total marks: 10 X 1= 15**

1. What does Cybersecurity protect?

a. Only physical infrastructure
b. Only software
c. Hardware, software, and data
d. Only company policies

Answer: (c)

Justification: Please refer to Slides 3 of Lecture 1 of Week 4

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2. What is the focus of Network Security?

a. Only securing physical buildings
b. Protecting network architecture from unauthorized access and improper use
c. Protecting only the company CEO's information
d. Solely focused on securing paper documents

Answer: (b)

Justification: Please refer to Slide 8 of Lecture 1 of Week 4

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3. What does the CIA Triad in Cybersecurity stand for?

a. Confidentiality, Integrity, Availability
b. Costing, Integrity, Assets
c. Company, Internet, Availability
d. Communication, Information, Assessment

Answer: (a)

Justification: Please refer to Slide 15 of Lecture 1 of Week 4

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4. What are the key differences between IIoT and traditional Automation?

a. No significant differences

b. Ubiquitous sensing, advanced analytics, IT tools, and methodologies
c. Only in terms of cost
d. Limited to the type of machinery used

Answer: (b)

Justification: Please refer to Slide 3 of Lecture 2 of Week 4

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5. What are the benefits of deploying sensors and actuators in IIoT?

a. Limited to academic purposes
b. To sell more software
c. To control, enhance, and optimize various functions
d. To reduce operational efficiency

Answer: (c)

Justification: Please refer to Slide 5 of Lecture 2 of Week 4

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6. How does IIoT benefit the mining industry?

a. By relaxing safety standards
b. Increasing disaster risks
c. Early disaster warning
d. Ignoring carbon footprint metrics

Answer: (c)

Justification: Please refer to Slide 19 of Lecture 2 of Week 4

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7. What is the Third Wave in GE's industrial level innovation?

a. The Industrial Intranet
b. The Digital Revolution
c. The Industrial Revolution
d. The Industrial Internet

Answer: (d)

Justification: Please refer to Slides 3 of Lecture 3 of Week 4

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8. Scarcity of non-renewable sources of energy pose a significant challenge with respect to energy consumption for the future smart industries and industrial internet?

a. True
b. False

Answer: (a)

Justification: Please refer to Page 83 of the book Introduction to Industrial Internet of Things and Industry 4.0 by S Misra, C Roy, A Mukherjee by CRC Press

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9. What was the negative effect of the Industrial Revolution?

a. Increased life expectancy
b. Environmental pollution
c. Rise in literacy
d. Technological advancement

Answer: (b)

Justification: Please refer to Slide 7 of Lecture 3 of Week 4

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10. Which of the following is not one of the three key elements of the Industrial Internet?

a. Intelligent machines
b. Advanced analytics
c. People at work
d. Social Networking

Answer: (d)

Justification: Please refer to Slide 10 of Lecture 3 of Week 4

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11. How is a smart sensor defined according to IEEE 1451 standard?

a. Sensor with large storage capacity
b. Sensor with standardized physical connection for communication
c. Sensor that operates independently
d. Sensor without the need for a power supply

Answer: (b)

Justification: Please refer to Slide 7 of Lecture 4 of Week 4

12. What is the purpose of the MRAA library in Intel IoT devices?

a. High-level gaming interface
b. Large data analysis
c. Cloud storage management
d. Low-level communication in GNU/Linux platform

Answer: (d)

Justification: Please refer to Slide 14 of Lecture 4 of Week 4

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13. What is a key aspect of the design philosophy for IIoT in industrial processes?

a. Facilitating communication between heterogeneous objects
b. Prioritizing manual control over automated systems
c. Limiting data interoperability
d. Reducing smart object autonomy

Answer: (a)

Justification: Please refer to Slide 6 of Lecture 5 of Week 4

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14. What is a feature of the Autonomous Pull Economy in Industrial Process 4.0?

a. High waste generation
b. Manual data processing
c. Limited resource optimization
d. End-to-end automation facility

Answer: (d)

Justification: Please refer to Slide 14 of Lecture 5 of Week 4

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15. Considering the future architecture of the smart Industrial plant, which among the following layers is responsible for autonomous decision making , machine learning and decision support system?

a. Connection Layer
b. Conversion Layer
c. Cyber Layer
d. Cognition Layer

Answer: (d)

Justification: Please refer to Page 190 of the book Introduction to Industrial Internet of Things and Industry 4.0 by S Misra, C Roy, A Mukherjee by CRC Press

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